

December 2025 CSCA Mathematics

1. If sets $A = \{1, 2\}$, $B = \{1, 2, 3\}$, then (...)

已知集合 $A=\{1,2\}$, $B=\{1,2,3\}$, 则 ()

A. $B \subseteq A$

B. $A = B$

C. $A \subseteq B$

D. $A \in B$

2. The solution set of the quadratic inequality $x^2 - 5x + 4 < 0$ is (...)

一元二次不等式 $x^2-5x+4<0$ 的解集为(...)

A. $(-\infty, 1) \cup (4, +\infty)$

B. $(1, 4)$

C. $(-\infty, -4) \cup (-1, +\infty)$

D. $(-4, -1)$

3. The domain of the function $y = \frac{1}{\sqrt{x+1}}$ is (...)

函数 $y = \frac{1}{\sqrt{x+1}}$ 的定义域是 (...)

A. $(-1, +\infty)$

B. $(-1, +\infty)$

C. $[-1, +\infty)$

D. $(0, +\infty)$

4. If $\{a_n\}$ is an arithmetic sequence with first term $a_1 = 2$, common difference $d = 3$, then $a_{100} = (..)$

已知 $\{a_n\}$ 是等差数列, 首项 $a_1 = 2$, 公差 $d=3$, 则 $a_{100} = (..)$

A. 302

B. 305

C. 299

D. 298

5. The point $(-1, 3 - \pi)$ is located in the ___ quadrant of the rectangular coordinate system.

点 $(-1, 3- \pi)$ 在平面直角坐标系中位于第___象限.

A. Second 第二象限

B. First 第一象限

C. Third 第三象限

D. Fourth 第四象限

6. $(\sin \frac{\pi}{4} + \cos \frac{\pi}{4}) \times \tan \frac{\pi}{4} = (...)$

A. $2\sqrt{2}$

B. 2

C. $\sqrt{2}$

D. 1

7. Which of the following functions is an odd function? ()

下列函数中, 为奇函数的是(...)

A. $y = x \sin x$

B. $y = \cos x$

C. $y = x^4$

D. $y = x^4$

8. The inverse function of $y = x^3$ is (...)

函数 $y=x^3$ 的反函数是 (...)

A. $y = \frac{x}{3}$

B. $y = 3x$

C. $y = x^3$

D. $y = \sqrt[3]{x}$

9. Which of the following functions is monotonically increasing? ()

下列函数中，单调递增的是 (...)

A. $y = \frac{1}{3}x$

B. $y = 2x + 1$

C. $y = \sin x$

D. $y = \log_{0.5} x$

10. Which point is located in the third quadrant? (...)

下列点中，位于第三象限的是(...)

A. (-1, 2)

B. (-1, -2)

C. (1, -2)

D. (1, 2)

11. The solution set of the fractional inequality $\frac{(x-1)}{(x-2)} \leq 1$ is (...)

分式不等式 $\frac{(x-1)}{(x-2)} \leq 1$ 的解集为 (...)

A. $(2, +\infty)$

B. $[1, 2]$

C. $(-\infty, 1]$

D. $(-\infty, 2)$

12. Given that a, b, and c form an arithmetic sequence and satisfy $a + c = 20$, then $b = ()$

已知 a, b 成等差数列，且满足 $a+c=20$ ，则 $b= (...)$

A. 0

B. 20

C. 10

D. 5

12. The angle of inclination of the line $y = \sqrt{3}x + 10$ is (...)

直线 $y=\sqrt{3}x+10$ 的倾斜角为 (...)

A. 30°

B. 45°

C. 60°

D. 120°

14. Which of the following inequalities is correct? (...)

下列不等式中，成立的是 (...)

A. $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$

B. $3^{2.25} > 3^3$

C. $2.1^{-2} > 1.2^{-2}$

D. $0.75^{-0.2} > 0.75^{-0.1}$

15. If point $P(1, y)$ lies on the terminal side of angle α , $\tan \alpha = 2$, then $y = (\dots)$

若点 $P(1, y)$ 在角 α 的终边上, 且 $\tan \alpha = 2$, 则 $y = (\dots)$

- A. 1
B. $\sqrt[3]{3}$
C. $\frac{1}{2}$
D. 2

16. Given that $\{a_n\}$ is an arithmetic sequence with $a_1 = 1$, $a_2 = 2$, then $a_{2025} = (\dots)$

已知 $\{a_n\}$ 是等差数列, $a_1 = 1$, $a_2 = 2$, 则 $a_{2025} = (\dots)$

- A. 20
B. 2
C. 2025
D. 2024

17. If a circle has center coordinates $(-3, 2)$ and radius 4, its equation is $(\dots)$

已知圆的圆心坐标为 $(-3, 2)$, 半径为 4, 则该圆的方程为 $(\dots)$

- A. $(x - 3)^2 + (y + 2)^2 = 16$
B. $(x - 3)^2 + (y + 2)^2 = 4$
C. $(x + 3)^2 + (y - 2)^2 = 16$
D. $(x + 3)^2 + (y - 2)^2 = 4$

18. The distance from point $P(-1, -1)$ to point $Q(1, 1)$ is $(\dots)$

点 $P(-1, -1)$ 到点 $Q(1, 1)$ 的距离为 $(\dots)$

- A. 4
B. $\sqrt{2}$
C. $2\sqrt{2}$
D. 2

19. The equation of the line passing through points $P(1, 2)$ and $Q(2, 4)$ is $(\dots)$

过点 $P(1, 2)$ 和 $Q(2, 4)$ 的直线方程为 $(\dots)$

- A. $y = 2x + 2$
B. $y = x + 2$
C. $y = 2x$
D. $y = 2x$

20. Given the equation of a circle $x^2 + y^2 + 2x - 2y = 0$, its radius is $(\dots)$

已知圆的方程为 $x^2 + y^2 + 2x - 2y = 0$, 则该圆的半径为 $(\dots)$

- A. 2
B. $\sqrt{2}$
C. $\frac{1}{2}$
D. 1

21. If α is an acute angle, and $\sin \alpha = \frac{\sqrt{3}}{2}$, then $\tan \alpha = (\dots)$

若 α 为锐角, 且 $\sin \alpha = \frac{\sqrt{3}}{2}$, 则 $\tan \alpha = (\dots)$

- $\frac{1}{2}$
 $\frac{1}{4}$
 $\sqrt{3}$
 $\frac{\sqrt{3}}{3}$

22. Given the first three terms of a geometric sequence $\{a_n\}$ are 1, 2, and 4, its general term formula is ()

已知等比数列 $\{a_n\}$ 的前三项为 1、2、4，则其通项公式为 ()

A. $a_n = 2n$

B. $a_n = 8$

C. $a_n = 2^{n-1}$

D. $a_n = 2^n$

23. Given $a > b$, then (...)

已知 $a > b$ ，则 (...)

A. $\frac{1}{a} < \frac{1}{b}$

B. $a^2 > b^2$

C. $5^a > 5^b$

D. $\sin a > \sin b$

24. The distance from point P(-1,2) to point Q(3,1) is (...)

点 P(-1,2)到点 Q(3,1)的距离为 (...)

A. $\sqrt{17}$

B. $\sqrt{13}$

C. $\sqrt{5}$

D. 5

25. Which of the following descriptions about the function $y = (\frac{1}{2})^x$ is correct? (...)

关于函数 $y = (\frac{1}{2})^x$ ，说法正确的是 (...)

A. The function is symmetric about the origin

B. The function is symmetric about the x-axis

函数关于原点对称

函数关于 x 轴对称

C. The function is monotonically increasing

D. The function is monotonically decreasing.

函数单调递增

函数单调递减

26. If $\sin a = \frac{1}{4}$, then $\cos 2a = (...)$

若 $\sin a = \frac{1}{4}$ ，则 $\cos 2a = (...)$

A. $-\frac{7}{8}$

B. $\frac{1}{8}$

C. $\frac{7}{8}$

D. $-\frac{1}{8}$

27. The intersection point of lines $l_1: y = 2x + 1$ and $l_2: x + y + 1 = 0$ is ().

直线 $l_1: y = 2x + 1$ 与直线 $l_2: x + y + 1 = 0$ 的交点坐标为 ()

A. $(\frac{1}{3}, \frac{2}{3})$

B. $(\frac{2}{3}, \frac{1}{3})$

C. $(\frac{-1}{3}, \frac{-2}{3})$

D. $(\frac{-2}{3}, \frac{-1}{3})$

28. The coordinates of the foci of the hyperbola $x^2 - y^2 = 1$ are (...)

双曲线 $x^2 - y^2 = 1$ 的焦点坐标为 (...)

- A. $(\pm \sqrt{2}, 0)$ B. $(0, \pm 2)$
C. $(0, \pm \sqrt{2})$ D. $(\pm 2, 0)$

29. $\log_2 32 + \log_{\left(\frac{1}{2}\right)} 32 = ()$

- A. -5 B. 10
C. 0 D. 5

30. Regarding the function $y = \cos x$, which statement is correct? (...)

关于函数 $y = \cos x$, 说法正确的是(...)

- A. $\cos(x + \pi) = \cos x$ B. This function is an odd function
该函数为奇函数
C. This function is monotonically increasing on the interval $[0, \pi]$. D. This function is a periodic function with period 2π .
该函数是以 2π 为周期的周期函数
该函数在区间 $[0, \pi]$ 上单调递增

31. Regarding the parabola $y^2 = 4x$, which statement is correct? (...)

关于抛物线 $y^2 = 4x$, 说法正确的是 (...)

- A. It passes through point $(-4, 4)$. B. Its axis of symmetry is $x = 0$.
C. Its focus coordinates are $(\pm 1, 0)$. D. Its focus coordinates are $(1, 0)$.

32. Which line is perpendicular to line 1: $x + y + 3 = 0$? (...)

下列直线中, 与直线 $l: x + y + 3 = 0$ 垂直的是 (...)

- A. $y = -2x + 1$ B. $y = x + 3$
C. $y = 1 - x$ D. $y = 2x + 1$

33. Given $\cos \alpha = -\frac{1}{2}$, $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, then $\sin\left(\frac{\alpha}{2}\right) = (\dots)$

已知 $\cos \alpha = -\frac{1}{2}$, $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, 则 $\sin\left(\frac{\alpha}{2}\right) = (\dots)$

- A. $-\frac{3}{2}$ B. $\frac{\sqrt{3}}{2}$
C. $\frac{1}{2}$ D. $-\frac{1}{2}$

34. Given $\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$, $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$, using the formula $\cos(\alpha - \beta) = \dots$,

we can get $\cos\left(\frac{\pi}{12}\right) = ()$

已知 $\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$, $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$, 利用公式 $\cos(\alpha - \beta) = \dots$, 可得 $\cos\left(\frac{\pi}{12}\right) = (\dots)$

A. $\frac{\sqrt{6}}{4}$

B. $\frac{\sqrt{6} + \sqrt{2}}{4}$

C. $\frac{\sqrt{6} - \sqrt{2}}{4}$

D. $\frac{(1 + \sqrt{3})}{4}$

35. Which function is identical to the function $y = |x|$? (...)

下列函数与函数 $y = |x|$ 是同一个函数的是 (...)

A. $y = \sqrt[3]{x^3}$

B. $y = (\sqrt{x})^2$

C. $y = x$

D. $y = \sqrt{x^2}$

36. If α is an acute angle, and $\cos \alpha = \frac{1}{2}$, then $\sin \frac{\alpha}{2} = (\dots)$

已知 α 为锐角, 且 $\cos \alpha = \frac{1}{2}$, 则 $\sin \frac{\alpha}{2} = (\dots)$

A. $\frac{\pm\sqrt{3}}{2}$

B. $\frac{\pm 1}{2}$

C. $\frac{\sqrt{3}}{2}$

D. $\frac{1}{2}$

37. Regarding reduction formulas, which equation is incorrect? ().

关于诱导公式, 下列灯饰错误

A. $\cos(2\pi - \alpha) = \cos \alpha$

B. $\tan(\pi + \alpha) = -\tan \alpha$

C. $\sin(\pi - \alpha) = \sin \alpha$

D. $\sin(\pi/2 - \alpha) = \cos \alpha$

38. The directrix equation of the parabola $y^2 = -x$ is (...)

抛物线 $y^2 = -x$ 的准线方程 (...)

A. $x = \frac{1}{2}$

B. $x = \frac{-1}{2}$

C. $x = \frac{1}{4}$

D. $x = \frac{-1}{4}$

39. Given $\frac{\cos \alpha - \sin \alpha}{\cos \alpha + \sin \alpha} = \frac{1}{3}$, then $\tan \alpha = ()$

A. $\frac{1}{2}$

B. $\frac{1}{4}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

40. If a sequence $\{a_n\}$ satisfies $a_1 = 1$, $a_n = \frac{1}{1+\frac{1}{a_{n-1}}}$ ($n \geq 2$), then $a_{1000} = (\dots)$

数列 $\{a_n\}$ 满足 $a_1 = 1$, $a_n = \frac{1}{1+\frac{1}{a_{n-1}}}$ ($n \geq 2$), 求 $a_{1000} = (\dots)$

A.50

B. $\frac{1}{100}$

C.1

D.100

41. Regarding the logarithmic function $y = \log_a x$, which statement is correct? (...)

关于对数函数 $y = \log_a x$, 下列描述正确 (...)

A. Its domain is $(-\infty, +\infty)$

定义域 $(-\infty, +\infty)$

B. When $a > 1$, it is monotonically decreasing on its domain

当 $a > 1$ 时, 在定义域上单调递减

C. When $a > 1$, for any $x > 0$, $\log_a x > 1$ always holds.

当 $a > 1$ 时, 对任意 $x > 0$, $\log_a x > 1$ 恒成立

D. When $0 < a < 1$, it is monotonically decreasing on its domain.

当 $0 < a < 1$ 时, 在定义域上单调递减

42. Regarding the ellipse $\frac{x^2}{4} + \frac{y^2}{20} = 1$ which statement is correct? (...)

对于椭圆 $\frac{x^2}{4} + \frac{y^2}{20} = 1$, 下列描述正确 (...)

A. Its two foci are $(0, \pm 4)$

焦点 $(0, \pm 4)$

B. Its two foci are $(\pm 4, 0)$.

焦点 $(\pm 4, 0)$

C. Its vertices are $(\pm 4, 0)$ and $(0, \pm 2\sqrt{5})$

D. 顶点 $(\pm 4, 0)$ 和 $(0, \pm 2\sqrt{5})$

D. The lengths of its major and minor axes are 20 and 4, respectively.

长轴和短轴的长度分别为 20 和 4.

43. A straight line l is perpendicular to the straight line $x + y + 4 = 0$ and passes through the intersection point of $x + y + 1 = 0$ and $3x + 2y - 1 = 0$. Then the equation of l is

若直线 l 与直线 $x + y + 4 = 0$ 垂直, 且经过直线 $x + y + 1 = 0$ 和 $3x + 2y - 1 = 0$ 的交点, 则 l 的方程是 (...)

A. $2x + y - 5 = 0$

B. $2x - y - 7 = 0$

C. $x - 2y - 1 = 0$

D. $x + 2y - 3 = 0$

44. Given O is the origin, A and B are two points in the plane, and P is the midpoint of AB , then the $\overrightarrow{OP} = (\dots)$

已知 O 为坐标原点, A, B 为平面内两点, P 为 AB 的中点, 则向量 $\overrightarrow{OP} = (\dots)$

A. $\frac{1}{2} (\overrightarrow{OA} + \overrightarrow{OB})$

B. $\overrightarrow{OA} + \overrightarrow{OB}$

C. $\overrightarrow{OB} - \overrightarrow{OA}$

D. $\frac{1}{2} (\overrightarrow{OA} - \overrightarrow{OB})$

45. If a complex number z satisfies $z^3 = 1$ and $z \neq 1$, then $1 + z + z^2 + \dots + z^{999} = (\dots)$

若复数 z 满足 $z^3 = 1$ 且 $z \neq 1$, 则 $1 + z + z^2 + \dots + z^{999} = (\dots)$

A. $\frac{1 \pm \sqrt{31}}{2}$

B. $\frac{-1 \pm \sqrt{31}}{2}$

C. 1

D. i

46. Given an arithmetic sequence $\{a_n\}$ satisfies $3(a_2 + a_6) + 2(a_6 + a_{10} + a_{14}) = 24$, then $S_{13} = (\dots)$

已知等差数列 $\{a_n\}$ 满足 $3(a_2 + a_6) + 2(a_6 + a_{10} + a_{14}) = 24$, 则 $S_{13} = (\dots)$

A. 104

B. 52

C. 39

D. 26

47. If the set $A = (-1, 3)$, $B = [2, 4]$, then $A \cap B$

若集合 $A = (-1, 3)$, $B = [2, 4]$, 求 $A \cap B$

A. $[2, 3]$

B. $(2, 3]$

C. $(2, 3)$

D. $(2, 3)$

48. The Chinese zodiac, also known as Shǔxiàng (属相), consists of twelve animals representing years in a cycle: Rat, Ox, Tiger, Rabbit, Dragon, Snake, Horse, Goat, Monkey, Rooster, Dog, and Pig. For example, people born in 1984, 1996, and 2008 are born in the Year of the Rat, while those born in 1985, 1997, and 2009 are born in the Year of the Ox, and so on. The probability that at least two out of three randomly selected people share the same zodiac sign is $(\dots)$

十二生肖(又称属相)是中国由十二种动物组成的纪年循环, 依次为鼠、牛、虎、兔、龙、蛇、马、羊、猴、鸡、狗、猪。例如, 1984、1996、2008 年出生的人属鼠, 1985、1997、2009 年出生的人属牛, 以此类推。随机选取三人, 其中至少两人生肖相同的概率为 $(\dots)$

A. $\frac{17}{72}$

B. $\frac{1}{12}$

C. $\frac{55}{72}$

D. $\frac{143}{144}$